azzouzi

azzouzi PROJECT PLANNING

PROJECT SPECIFICATION

E-commerce Website

PREPARED FOR

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PROJECT TYPE

E-commerce Website

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Prepared by ProjectMind AI

1. Project Presentation

This section introduces the project, its main objective, its purpose and the expected outcomes.

General Presentation

Hwayji is a proposed e-commerce website intended to let customers discover and purchase products online. The client describes a broad product offering and a goal of making everyday shopping simple, convenient, and accessible. The requested scope names a product catalog, shopping cart, order management, and an administrator dashboard. Specific product categories, operating markets, and fulfillment arrangements have not been provided.

Main Objective

Build an online shopping platform where customers can browse a product catalog, compare product prices, add selected products to a cart, and place orders. Administrators should be able to use an administrative dashboard to manage the platform's products and orders, subject to confirmation of the required administrative tasks. The client also expects a smooth experience, secure transactions, reliable service, and fast delivery; the payment and delivery processes that would support those expectations remain to be specified.

Problem Statement

Customers need a convenient way to discover and order products online. Hwayji aims to bring product discovery, price comparison, cart-based purchasing, and order handling into one website. The operational details required to provide secure payment, reliable order fulfillment, and fast delivery have not yet been defined.

Overall Project Workflow

A visitor opens the website and browses the available catalog. They review product information and prices, add items to a shopping cart, and proceed toward placing an order. The precise checkout steps, account requirements, payment method, delivery options, and order notifications are to be specified. After an order is submitted, an administrator uses the dashboard to manage it according to agreed order-processing rules. The client has not specified inventory processes, fulfillment responsibilities, returns, or customer support workflows.

Client Requirements

- Provide an e-commerce website for selling products online.
- Provide a product catalog.
- Provide a shopping cart.
- Provide order management.
- Provide an administrative dashboard.
- Support a user-friendly shopping experience and allow customers to explore products, compare prices, and place orders.
- Meet the stated expectations of secure transactions, reliable service, and fast delivery; implementation details and service-level definitions are to be specified.
- Use the client-provided colors #FFFFFF, #000000, and #07AB3B as visual references.

Expected Results

- A responsive website through which customers can browse products and initiate purchases.
- A cart-based purchasing flow whose checkout, payment, and delivery details are confirmed before implementation.
- An administrative interface for the agreed product and order management tasks.
- A consistent visual identity based on the supplied color references.
- A tested and deployable application, subject to confirmation of hosting, operational ownership, and launch requirements.

2. Context & Environment

This section describes the business context, target audience, users and project constraints.

Business Context

The project concerns online retail. Hwayji is presented as a platform offering a wide range of products at competitive prices, but the actual product categories, suppliers, stock ownership model, sales territory, and business operating model have not been specified.

Project Environment

The requested product is an e-commerce website with customer-facing shopping functionality and an administrative dashboard. No existing website, product-management system, payment provider, delivery provider, inventory system, or other integration has been identified. The deployment environment and operational support model are to be specified.

Target Audience

Online shoppers seeking to discover and purchase products conveniently. The client has not specified geographic markets, supported languages, accessibility requirements, customer demographics, or whether business customers are also in scope.

Application Users

Customer or visitor

The intended audience is online shoppers. Visitors are expected to explore products and use the shopping flow. Whether customers must create accounts or can place guest orders is to be specified.

Permissions:

Browse and view products; this is consistent with the client-described catalog., Compare product prices; this is stated as an intended shopping activity., Add products to and manage items in a shopping cart; the cart is explicitly requested., Place an order through the agreed checkout process; placing orders is described by the client, while checkout rules are to be specified., View order history or account information only if these capabilities are approved; they were not specified by the client.

Administrator

An authorized operator using the explicitly requested admin dashboard. The specific business responsibilities and number of administrator permission levels are to be specified.

Permissions:

Access the admin dashboard., Manage orders within the scope and workflow approved by the client., Manage catalog entries only if product administration is confirmed as part of the dashboard scope; the client requested a catalog and dashboard but did not detail catalog administration., Manage users, inventory, promotions, or site settings only if separately approved; these functions were not specified.

Project Constraints

- The client supplied #FFFFFF, #000000, and #07AB3B as color references. Their intended use and any existing brand guidelines are to be confirmed.
- The client expects secure transactions, but no payment provider, payment methods, or payment-flow requirements were specified.
- Fast delivery and reliable service are stated expectations, but delivery coverage, delivery partners, shipping rules, and operational commitments are to be specified.
- Product categories, catalog size, product variants, stock management, and product data ownership are to be specified.
- Order statuses, cancellation rules, returns, refunds, and customer notifications are to be specified.
- Supported languages, currencies, taxes, legal jurisdictions, and applicable consumer-protection requirements are to be specified.

- Hosting ownership, domain, deployment environment, backup policy, and post-launch support arrangements are to be specified.

Information to Be Confirmed

- Which product categories and initial catalog size will the website support?
- Who supplies and maintains product descriptions, images, prices, variants, and availability?
- Does the administrator dashboard need catalog creation and editing, inventory management, reporting, or only order management?
- What order statuses and operational steps should administrators follow?
- Is customer registration required, or should guest checkout be supported?
- Which payment methods and payment provider should be supported? Is payment collected online, on delivery, or through another process?
- Which regions will be served, and what delivery methods, fees, delivery estimates, and providers apply?
- What cancellation, returns, exchange, refund, and customer support policies should be represented?
- Which languages and currencies are required?
- What tax, invoicing, privacy, and consumer-protection requirements apply in the target markets?
- Are promotional codes, discounts, product reviews, wish lists, or product search and filtering required?
- Are email, SMS, or other order notifications required, and which service should send them?
- Are there existing brand assets, a logo, typography rules, or reference websites in addition to the supplied colors?
- What are the expected launch environment, domain, hosting ownership, backup needs, and post-launch maintenance arrangements?

3. Mockups & Visual Design

This section defines the proposed visual direction, interface structure and user experience recommendations.

Art Direction

Recommendation: create a clear, approachable retail identity that makes product discovery and purchase actions easy to understand. Use the supplied green as an accent for primary actions and status indicators, with white and black for the main interface. This is a design proposal, not a description of existing mockups.

Graphic Style

Recommendation: use a clean, product-led layout with legible typography, consistent spacing, restrained decoration, and prominent product photography. Maintain clear visual separation between customer shopping pages and the administrative dashboard. No mockups or existing visual references beyond the client's color references were provided.

Color Palette

- color: #FFFFFF, source: Client-provided visual reference, recommendedUse: Primary page and card backgrounds.
- color: #000000, source: Client-provided visual reference, recommendedUse: Primary text and selected high-contrast interface elements.
- color: #07AB3B, source: Client-provided visual reference, recommendedUse: Accent color for primary actions, selected states, and positive status indicators, subject to contrast testing.
- color: Neutral gray shades, source: AI recommendation, recommendedUse: Supporting text, borders, dividers, and inactive controls. Exact shades should be selected during design and checked for accessibility.

Visual References

- The client supplied the color references #FFFFFF, #000000, and #07AB3B.
- No website references, logo files, brand guidelines, or mockups were supplied.

Main Interfaces

Home page

Recommended entry point for discovering the store and reaching the catalog. Specific promotional content has not been requested.

- Brand header and primary navigation
- Catalog entry point
- Search entry point if approved
- Product or category highlights based on client-provided content
- Cart access
- Footer with confirmed business and policy information

Product catalog and product detail

Customer-facing screens for browsing products and reviewing product information and prices. Catalog organization and filtering requirements are to be confirmed.

- Product image, name, and price
- Product detail information supplied by the client
- Category navigation or filtering if approved
- Add-to-cart action
- Availability information if stock management is included

Shopping cart

Screen for reviewing and adjusting items selected for purchase.

- Cart item summaries and prices
- Quantity adjustment and item removal
- Subtotal and any confirmed charges
- Proceed-to-checkout action
- Clear empty-cart state

Checkout and order confirmation

Recommended purchase flow; payment, delivery, account, and confirmation rules must be agreed before detailed design.

- Customer and delivery details as required by the approved process
- Order summary
- Payment selection or instructions for the confirmed payment method
- Final order submission
- Order confirmation and reference

Admin dashboard

Administrative interface explicitly requested by the client. The exact dashboard modules and administrator workflow are to be confirmed.

- Secure administrator access
- Order list and order detail views
- Order status actions based on agreed workflow
- Catalog management only if approved
- Clear navigation and validation feedback

UX Recommendations

- Make catalog browsing, cart access, and the next purchase action easy to locate on both desktop and mobile.
- Keep checkout steps concise and disclose only confirmed charges and delivery information before order submission.
- Provide clear feedback when items are added to the cart, quantities change, or an order is submitted.
- Use accessible text contrast and do not communicate order or validation states through color alone.

- Design useful loading, empty, error, and unavailable-product states.
- Use consistent product image proportions and request appropriately sized image assets from the client.
- Confirm whether guest checkout is allowed before designing account-related steps.

4. Technical Specifications

This section presents the functional requirements, proposed architecture, technical constraints and estimated development schedule.

Functional Requirements

Product catalog High

Present products for customer exploration, including product information and prices. The client explicitly requested a catalog and described product exploration and price comparison. Categories, search, filters, variants, stock display, and catalog administration require confirmation.

Source: Client requirement

Shopping cart High

Allow customers to add products to a cart and review selected items before placing an order. Quantity editing, cart persistence, and guest-versus-account behavior need confirmation.

Source: Client requirement

Order placement High

Support the customer activity of placing an order described in the client presentation. Checkout fields, payment behavior, delivery selection, and order confirmation rules are to be specified.

Source: Client requirement

Order management High

Provide an administrative process to view and manage orders through the requested dashboard. Order statuses, permitted actions, cancellation handling, and fulfillment workflow are to be specified.

Source: Client requirement

Admin dashboard High

Provide an administrative interface. The client requested the dashboard but did not define its modules, user permissions, reporting, or catalog-management scope.

Source: Client requirement

Secure online payment integration High

Integrate a payment provider only after the client confirms whether online payment is required, the provider, supported methods, target markets, and payment lifecycle. The client stated secure transactions as an expectation but did not specify an implementation.

Source: AI recommendation

Administrator authentication and authorization High

Protect dashboard access with authenticated administrator accounts and role-based authorization appropriate to the agreed operator roles.

Source: AI recommendation

Catalog search and filtering Medium

Consider search, category navigation, and filters to help customers explore a potentially broad product range. Their exact inclusion depends on catalog size and client approval.

Source: AI recommendation

Order notifications Medium

Consider sending customers and administrators order-related notifications through an approved email or messaging service. Notification events, templates, and provider are to be specified.

Source: AI recommendation

Inventory availability controls Medium

Consider stock tracking and preventing orders for unavailable items if Hwayji owns or manages inventory. The inventory model has not been provided.

Source: AI recommendation

Proposed Technical Architecture

Frontend: Recommendation: build a responsive storefront and admin interface using a maintained component-based framework such as React with TypeScript. Use accessible, reusable components and keep customer and administrator navigation clearly separated. The final framework should be selected after confirming team expertise, hosting constraints, SEO requirements, and expected catalog scale.

Backend: Recommendation: implement a server-side application exposing a documented REST API for catalog, cart, checkout, orders, and administrative operations. Enforce authorization and business rules on the server rather than relying on frontend checks. Select the backend framework after confirming the delivery team's capabilities and integration requirements.

Database: Recommendation: use a relational database such as PostgreSQL for product records, order records, and other structured transactional data. Define the data model after confirming product variants, inventory ownership, payment lifecycle, address handling, and order status rules. Store images in managed object storage rather than as database binary content.

Hosting: Recommendation: use a managed cloud or platform hosting service with separate development, staging, and production environments; automated deployment; encrypted transport; managed database backups; and centralized application monitoring. Provider, region, domain, data residency, and operational ownership are to be specified.

Architecture Justification

A typed, component-based frontend and an API-backed service are suitable for separating the customer storefront from administration while supporting a maintainable catalog and order workflow. A relational database fits structured commerce records and transactional consistency. Managed hosting and storage reduce routine infrastructure administration, but the final choices depend on confirmed scale, integrations, geographic requirements, and team capabilities.

Security Requirements

- Serve the site and APIs over HTTPS and apply secure transport settings in production.
- Require authenticated access to administrative functions and enforce server-side authorization for every protected operation.
- Use a maintained authentication approach, secure password handling if passwords are used, and account recovery controls as appropriate to the approved account model.
- Validate and normalize user input on the server, use parameterized database access, and protect against common web vulnerabilities including injection and cross-site scripting.
- Use CSRF protections where applicable, secure session or token handling, and rate limiting for sensitive endpoints.
- Do not store raw payment-card data. If online payments are approved, use a suitable payment provider and follow its security and integration requirements.
- Restrict administrative access and sensitive data exposure according to approved roles; log relevant administrative and order-management actions.
- Apply data minimization, access controls, backup protection, and retention policies aligned with the confirmed jurisdictions and business needs.
- Keep application dependencies and deployment environments maintained, and define a process for addressing security updates and incidents.

Performance Requirements

- Optimize product images using appropriate formats, compression, responsive sizes, and lazy loading where suitable.
- Use pagination or incremental loading for large catalogs and administrative order lists.
- Add appropriate database indexes and review queries against the confirmed catalog and order volumes.
- Set performance targets after expected traffic, catalog size, hosting region, and business needs are confirmed.
- Monitor application errors, API response times, and availability after launch.

Responsive Design & Compatibility

Use a responsive layout suitable for mobile phones, tablets, and desktop screens; responsive behavior is recommended because the product is intended for online shoppers, although device targets were not specified., Test current stable versions of major browsers, including Chrome, Safari, Firefox, and Edge, once the supported browser policy is agreed., Support keyboard navigation, visible focus states, semantic structure, and accessible contrast as baseline usability and accessibility recommendations., Confirm supported operating systems, minimum browser versions, and any accessibility compliance target with the client.

External Integrations

- Payment provider: not specified. Confirm whether online payment is required, supported methods, target countries, and preferred provider before implementation.
- Delivery or shipping provider: not specified. Confirm service coverage, integration availability, delivery rules, and whether tracking is required.
- Email or SMS messaging: not specified. Consider an approved provider for order notifications if notifications are required.
- Analytics or marketing services: not specified and not assumed to be part of the initial scope.
- No external APIs or existing business-system integrations were identified in the supplied project information.

UML Diagrams

UML DIAGRAM	PURPOSE / DESCRIPTION	MAIN ELEMENTS
Order lifecycle sequence (conceptual)	A conceptual view of the core purchase flow and its administrative handoff. Payment and delivery interactions are conditional and require client confirmation; this is a proposed analysis diagram, not a confirmed implementation specification.	Customer browses catalog, Customer adds products to cart, Customer submits order details through the approved checkout, Store backend validates the order and records it, Payment provider interaction, if online payment is approved, Administrator reviews and manages the order, Delivery or fulfillment process, according to the confirmed operating model, Customer receives confirmation or status updates if notifications are approved

Development Duration Estimation

ESTIMATED DEVELOPMENT DURATION

Approximately 10.5 to 15 weeks.

1. Requirements analysis and planning	1 to 1.5 weeks

- Confirm product, catalog, cart, checkout, order, and dashboard scope.
- Clarify payment, delivery, inventory, account, language, currency, and jurisdiction requirements.
- Document acceptance criteria, operational workflows, and integration dependencies.

2. UI/UX design — **1.5 to 2 weeks**

- Define customer and administrator journeys.
- Produce responsive wireframes and visual designs for agreed key screens.
- Apply the supplied color references and review accessibility and usability.
- Obtain client feedback and approval before implementation.

3. Backend and database development — **2.5 to 3.5 weeks**

- Define and implement the agreed product, cart, and order data model.
- Build API endpoints and server-side business validation.
- Implement administrator authentication and authorization.
- Implement the approved order-management workflow.

4. Frontend development — **2.5 to 3.5 weeks**

- Build the responsive storefront and catalog interfaces.
- Implement cart interactions and the approved checkout experience.
- Build dashboard views for the confirmed administrative tasks.
- Connect interfaces to the backend API and handle loading, empty, and error states.

5. Database and API integration — **1 to 1.5 weeks**

- Integrate the application with the selected database and verify data flows.
- Integrate confirmed external services, if any, such as payment or messaging providers.
- Validate failure handling and data consistency across the order flow.

6. Testing and bug fixing — **1.5 to 2 weeks**

- Test catalog, cart, checkout, order handling, and dashboard workflows.
- Perform responsive, browser compatibility, security, and regression checks.
- Resolve defects and support client acceptance testing.

7. Deployment and delivery — **0.5 to 1 week**

- Configure agreed production hosting, domain, and deployment settings.
- Run production-readiness checks and verify backup and monitoring configuration.
- Deploy the approved release and provide agreed handover documentation.

Estimation Justification

This is a preliminary technical estimate for a custom e-commerce website with a catalog, cart, order workflow, and administrative dashboard. The range reflects uncertainty around product administration, customer accounts, payment and delivery integrations, inventory, content preparation, and approval cycles. Phase estimates are planning ranges and may involve some parallel work; the total is an indicative delivery duration rather than a committed deadline. The estimate may change after requirements validation and design approval.

Factors That May Influence the Timeline

- Number of products, categories, variants, and amount of catalog data to be prepared or imported.
- Whether guest checkout, customer accounts, order history, and account recovery are required.
- Payment-provider selection, onboarding, testing requirements, and supported payment methods.
- Delivery-provider integration, shipping rules, tracking, and service-area complexity.
- Inventory ownership, stock synchronization, and product availability rules.
- Required languages, currencies, taxes, invoicing, and jurisdiction-specific compliance.
- Complexity of administrator roles, reports, catalog operations, and order status workflows.
- Availability and quality of product images, descriptions, brand assets, and client feedback.
- Hosting, data residency, security, accessibility, and operational requirements.

Technical Assumptions

- The initial product is a responsive web application; native mobile applications were not requested.
- The client will provide product data, images, brand assets, business policies, and access to any approved external services.
- Payment, delivery, notifications, customer accounts, inventory, and catalog administration are not considered confirmed integrations or features until the client specifies them.
- The administrative dashboard will require protected access; administrator identity management and permission levels must be agreed.
- The application will use a supported, maintained technology stack and managed production services, subject to client approval.
- No existing platform migration, legacy-system integration, data import, or third-party API dependency was included in the estimate because none was identified.
- Legal, privacy, tax, and consumer-protection requirements depend on the markets in which Hwayji will operate and must be confirmed before launch.

> **Planning Note**
>
> The development duration is an initial estimate based on the information currently available. The final schedule may change after detailed requirements validation, design approval and technical assessment.